(9.9) $\vec{E}(\vec{r},t) = E_0 \cos(\vec{k}\cdot\vec{r} - \omega t + \delta)\,\hat{n}$

$\vec{B}(\vec{r},t) = \frac{1}{c} E_0 \cos(\vec{k}\cdot\vec{r} - \omega t + \delta)\,(\hat{k}\times\hat{n})$

always $k = |\vec{k}| = \frac{\omega}{c}$

$\delta = 0$

(a) $\vec{k} = -\frac{\omega}{c}\hat{x} \Rightarrow \vec{k}\cdot\vec{r} = -\frac{\omega}{c}x$

$\hat{n} = \hat{z} \Rightarrow \hat{k}\times\hat{n} = -\hat{x}\times\hat{z} = \hat{y}$

$$\Rightarrow \vec{E} = E_0 \cos\left(\frac{\omega}{c}x + \omega t\right)\hat{z}$$

$$\vec{B} = \frac{1}{c} E_0 \cos\left(\frac{\omega}{c}x + \omega t\right)\hat{y}$$

(b) $\vec{k} = \frac{\omega}{c}\,\frac{\hat{x}+\hat{y}+\hat{z}}{\sqrt{3}}$ so that $\hat{k} = \frac{1}{\sqrt{3}}(1,1,1)$ is unit vector.

Now $\hat{n} \in xz$ plane $\Rightarrow \hat{n} = \alpha\hat{x} + \beta\hat{z}$

But need to make sure that $\hat{k}\cdot\hat{n} = 0 \Rightarrow \alpha = -\beta$

$\Rightarrow$ normalizing $\hat{n} = \frac{1}{\sqrt{2}}(\hat{x} - \hat{z})$

Then $\hat{k}\times\hat{n} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{vmatrix} = \hat{x}\left(-\frac{1}{\sqrt{6}}\right) - \hat{y}\left(-\frac{1}{\sqrt{6}} - \frac{1}{\sqrt{6}}\right) + \hat{z}\left(-\frac{1}{\sqrt{6}}\right) =$

$$= \frac{-1}{\sqrt{6}}\left(\hat{x} - 2\hat{y} + \hat{z}\right)$$

$$\vec{E} = E_0 \cos\left(\frac{\omega}{\sqrt{3}c}(x+y+z) - \omega t\right)\frac{\hat{x}-\hat{z}}{\sqrt{2}}$$

$$\vec{B} = \frac{1}{c} E_0 \cos\left(\frac{\omega}{\sqrt{3}c}(x+y+z) - \omega t\right)\frac{-\hat{x}+2\hat{y}-\hat{z}}{\sqrt{6}}$$

9.15 $Ae^{iax} + Be^{ibx} = Ce^{icx}$

let $x=0$: $A+B=C$.

Take derivative with respect to x of both sides:

$$iaAe^{iax} + ibBe^{ibx} = icCe^{icx}$$

Set $x=0$:

$$aA + bB = cC$$

Taking second derivative at $x=0$ will give:

$$a^2A + b^2B = c^2C$$

$$\Rightarrow \begin{cases} C = A+B \\ C = \frac{a}{c}A + \frac{b}{c}B \\ C = \left(\frac{a}{c}\right)^2 A + \left(\frac{b}{c}\right)^2 B \end{cases} \quad A\left(1-\frac{a}{c}\right) = \left(\frac{b}{c}-1\right)B$$

$$\Rightarrow A+B = \left(\frac{a}{c}\right)^2 A + \left(\frac{b}{c}\right)^2 B$$

$$A\left(1-\frac{a^2}{c^2}\right) = \left(\frac{b^2}{c^2}-1\right)B$$

$$\frac{\left(\frac{b}{c}-1\right)}{1-\frac{a}{c}}\left(1-\frac{a^2}{c^2}\right)B = \left(\frac{b^2}{c^2}-1\right)B$$

$$\Rightarrow 1+\frac{a}{c} = \frac{b}{c}+1 \Rightarrow \underline{a=b}$$

$$\Rightarrow A\left(1-\frac{a}{c}\right) = A\left(1-\frac{b}{c}\right) = \left(\frac{b}{c}-1\right)B \Rightarrow \underline{A=-B}$$

which is not true

$$\Rightarrow 1-\frac{b}{c} = \frac{b}{c}-1 = 0 \Rightarrow \underline{b=c}$$

$$\Rightarrow a=b=c \text{ and } A+B=C.$$

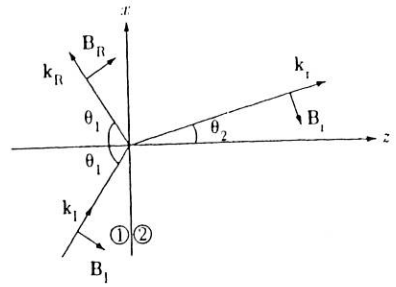
9.16

Problem 9.16

$$\left\{\begin{aligned}\tilde{\mathbf{E}}_I &= \tilde{E}_{0_I} e^{i(\mathbf{k}_I\cdot\mathbf{r}-\omega t)}\hat{\mathbf{y}},\\ \tilde{\mathbf{B}}_I &= \frac{1}{v_1}\tilde{E}_{0_I} e^{i(\mathbf{k}_I\cdot\mathbf{r}-\omega t)}(-\cos\theta_1\,\hat{\mathbf{x}}+\sin\theta_1\,\hat{\mathbf{z}});\end{aligned}\right\}$$

$$\left\{\begin{aligned}\tilde{\mathbf{E}}_R &= \tilde{E}_{0_R} e^{i(\mathbf{k}_R\cdot\mathbf{r}-\omega t)}\hat{\mathbf{y}},\\ \tilde{\mathbf{B}}_R &= \frac{1}{v_1}\tilde{E}_{0_R} e^{i(\mathbf{k}_R\cdot\mathbf{r}-\omega t)}(\cos\theta_1\,\hat{\mathbf{x}}+\sin\theta_1\,\hat{\mathbf{z}});\end{aligned}\right\}$$

$$\left\{\begin{aligned}\tilde{\mathbf{E}}_T &= \tilde{E}_{0_T} e^{i(\mathbf{k}_T\cdot\mathbf{r}-\omega t)}\hat{\mathbf{y}},\\ \tilde{\mathbf{B}}_T &= \frac{1}{v_2}\tilde{E}_{0_T} e^{i(\mathbf{k}_T\cdot\mathbf{r}-\omega t)}(-\cos\theta_2\,\hat{\mathbf{x}}+\sin\theta_1\,\hat{\mathbf{z}});\end{aligned}\right\}$$



Boundary conditions: $\left\{\begin{array}{ll}\text{(i) } \epsilon_1 E_1^{\perp} = \epsilon_2 E_2^{\perp}, & \text{(iii) } \mathbf{E}_1^{\parallel} = \mathbf{E}_2^{\parallel},\\ \text{(ii) } B_1^{\perp} = B_2^{\perp}, & \text{(iv) } \frac{1}{\mu_1}\mathbf{B}_1^{\parallel} = \frac{1}{\mu_2}\mathbf{B}_2^{\parallel}.\end{array}\right.$

Law of refraction: $\dfrac{\sin\theta_2}{\sin\theta_1} = \dfrac{v_2}{v_1}$. [Note: $\mathbf{k}_I\cdot\mathbf{r}-\omega t = \mathbf{k}_R\cdot\mathbf{r}-\omega t = \mathbf{k}_T\cdot\mathbf{r}-\omega t$, at $z=0$, so we can drop all exponential factors in applying the boundary conditions.]

Boundary condition (i): $0=0$ (trivial). Boundary condition (iii): $\boxed{\tilde{E}_{0_I}+\tilde{E}_{0_R}=\tilde{E}_{0_T}.}$

Boundary condition (ii): $\dfrac{1}{v_1}\tilde{E}_{0_I}\sin\theta_1 + \dfrac{1}{v_1}\tilde{E}_{0_R}\sin\theta_1 = \dfrac{1}{v_2}\tilde{E}_{0_T}\sin\theta_2 \Rightarrow \tilde{E}_{0_I}+\tilde{E}_{0_R} = \left(\dfrac{v_1\sin\theta_2}{v_2\sin\theta_1}\right)\tilde{E}_{0_T}$.

But the term in parentheses is 1, by the law of refraction, so this is the same as (ii).

Boundary condition (iv): $\dfrac{1}{\mu_1}\left[\dfrac{1}{v_1}\tilde{E}_{0_I}(-\cos\theta_1) + \dfrac{1}{v_1}\tilde{E}_{0_R}\cos\theta_1\right] = \dfrac{1}{\mu_2 v_2}\tilde{E}_{0_T}(-\cos\theta_2) \Rightarrow$

$\tilde{E}_{0_I}-\tilde{E}_{0_R} = \left(\dfrac{\mu_1 v_1\cos\theta_2}{\mu_2 v_2\cos\theta_1}\right)\tilde{E}_{0_T}$. Let $\boxed{\alpha \equiv \dfrac{\cos\theta_2}{\cos\theta_1};\ \beta\equiv\dfrac{\mu_1 v_1}{\mu_2 v_2}.}$ Then $\boxed{\tilde{E}_{0_I}-\tilde{E}_{0_R}=\alpha\beta\tilde{E}_{0_T}.}$

Solving for $\tilde{E}_{0_R}$ and $\tilde{E}_{0_T}$: $2\tilde{E}_{0_I} = (1+\alpha\beta)\tilde{E}_{0_T} \Rightarrow \tilde{E}_{0_T} = \left(\dfrac{2}{1+\alpha\beta}\right)\tilde{E}_{0_I}$;

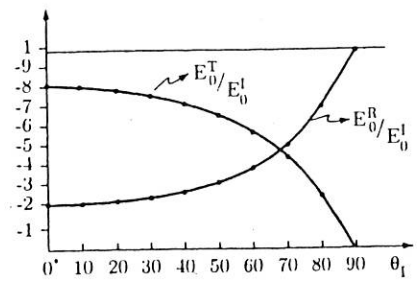
$\tilde{E}_{0_R} = \tilde{E}_{0_T}-\tilde{E}_{0_I} = \left(\dfrac{2}{1+\alpha\beta}-\dfrac{1+\alpha\beta}{1+\alpha\beta}\right)\tilde{E}_{0_I} \Rightarrow \tilde{E}_{0_R} = \left(\dfrac{1-\alpha\beta}{1+\alpha\beta}\right)\tilde{E}_{0_I}$.

Since α and β are positive, it follows that $2/(1+\alpha\beta)$ is positive, and hence the *transmitted* wave is *in phase* with the incident wave, and the (real) amplitudes are related by $\boxed{E_{0_T} = \left(\dfrac{2}{1+\alpha\beta}\right)E_{0_I}.}$ The *reflected* wave is

in phase if $\alpha\beta < 1$ and 180° out of phase if $\alpha\beta < 1$; the (real) amplitudes are related by $\boxed{E_{0_R} = \left|\frac{1-\alpha\beta}{1+\alpha\beta}\right| E_{0_I}.}$

These are the **Fresnel equations** for polarization perpendicular to the plane of incidence.

To construct the graphs, note that $\alpha\beta = \beta\frac{\sqrt{1-\sin^2\theta/\beta^2}}{\cos\theta} = \frac{\sqrt{\beta^2-\sin^2\theta}}{\cos\theta}$, where θ is the angle of incidence, so, for $\beta = 1.5$, $\alpha\beta = \frac{\sqrt{2.25-\sin^2\theta}}{\cos\theta}$.



Is there a Brewster's angle? Well, $E_{0_R} = 0$ would mean that $\alpha\beta = 1$, and hence that

$$\alpha = \frac{\sqrt{1-(v_2/v_1)^2\sin^2\theta}}{\cos\theta} = \frac{1}{\beta} = \frac{\mu_2 v_2}{\mu_1 v_1}, \text{ or } 1-\left(\frac{v_2}{v_1}\right)^2\sin^2\theta = \left(\frac{\mu_2 v_2}{\mu_1 v_1}\right)^2\cos^2\theta, \text{ so}$$

$1 = \left(\frac{v_2}{v_1}\right)^2\left[\sin^2\theta + (\mu_2/\mu_1)^2\cos^2\theta\right]$. Since $\mu_1 \approx \mu_2$, this means $1 \approx (v_2/v_1)^2$, which is only true for optically indistinguishable media, in which case there is of *course* no reflection—but that would be true at *any* angle, not just at a special "Brewster's angle". [If μ_2 were substantially different from μ_1, and the relative velocities were just right, it *would* be possible to get a Brewster's angle for this case, at

$$\left(\frac{v_1}{v_2}\right)^2 = 1-\cos^2\theta + \left(\frac{\mu_2}{\mu_1}\right)^2\cos^2\theta \Rightarrow \cos^2\theta = \frac{(v_1/v_2)^2-1}{(\mu_2/\mu_1)^2-1} = \frac{(\mu_2\epsilon_2/\mu_1\epsilon_1)-1}{(\mu_2/\mu_1)^2-1} = \frac{(\epsilon_2/\epsilon_1)-(\mu_1/\mu_2)}{(\mu_2/\mu_1)-(\mu_1/\mu_2)}.$$

But the media would be very peculiar.]

By the same token, δ_R is either always 0, or always π, for a given interface—it does not switch over as you change θ, the way it does for polarization in the plane of incidence. In particular, if $\beta = 3/2$, then $\alpha\beta > 1$, for

$$\alpha\beta = \frac{\sqrt{2.25-\sin^2\theta}}{\cos\theta} > 1 \text{ if } 2.25-\sin^2\theta > \cos^2\theta, \text{ or } 2.25 > \sin^2\theta+\cos^2\theta = 1. \checkmark$$

In general, for $\beta > 1$, $\alpha\beta > 1$, and hence $\delta_R = \pi$. For $\beta < 1$, $\alpha\beta < 1$, and $\delta_R = 0$.

At normal incidence, $\alpha = 1$, so Fresnel's equations reduce to $E_{0_T} = \left(\frac{2}{1+\beta}\right)E_{0_I}$; $E_{0_R} = \left|\frac{1-\beta}{1+\beta}\right|E_{0_I}$, consistent with Eq. 9.82.

Reflection and Transmission coefficients: $\boxed{R = \left(\frac{E_{0_R}}{E_{0_I}}\right)^2 = \left(\frac{1-\alpha\beta}{1+\alpha\beta}\right)^2.}$ Referring to Eq. 9.116,

$$T = \frac{\epsilon_2 v_2}{\epsilon_1 v_1}\alpha\left(\frac{E_{0_T}}{E_{0_I}}\right)^2 = \boxed{\alpha\beta\left(\frac{2}{1+\alpha\beta}\right)^2.}$$

$$R+T = \frac{(1-\alpha\beta)^2+4\alpha\beta}{(1+\alpha\beta)^2} = \frac{1-2\alpha\beta+\alpha^2\beta^2+4\alpha\beta}{(1+\alpha\beta)^2} = \frac{(1+\alpha\beta)^2}{(1+\alpha\beta)^2} = 1. \checkmark$$

9.19

Start from eq. 9.126

$$\varkappa = \omega\sqrt{\frac{\varepsilon\mu}{2}}\left[\sqrt{1+\left(\frac{\sigma}{\varepsilon\omega}\right)^2}-1\right]^{1/2}$$

(a) $\sigma \ll \omega\varepsilon \Rightarrow$ expand square root $\sqrt{1+\left(\frac{\sigma}{\varepsilon\omega}\right)^2}-1 \approx \frac{1}{2}\left(\frac{\sigma}{\varepsilon\omega}\right)^2$

$$\varkappa = \omega\sqrt{\frac{\varepsilon\mu}{2}}\sqrt{\frac{1}{2}\left(\frac{\sigma}{\varepsilon\omega}\right)^2} = \frac{1}{2}\sigma\sqrt{\frac{\mu}{\varepsilon}} \Rightarrow d = \frac{1}{\varkappa} = \frac{2}{\sigma}\sqrt{\frac{\varepsilon}{\mu}}$$

For water $\varepsilon = \varepsilon_r\varepsilon_0 \cong 80\,\varepsilon_0$ (p. 180) $\quad \varepsilon_0 = 8.85\cdot 10^{-12}$

$\mu = \mu_0(1+\chi_m) \approx \mu_0$ (p. 275) $\quad \mu_0 = 4\pi\cdot 10^{-7}$

$\sigma = \frac{1}{\rho} = \frac{1}{2.5\cdot 10^5}$ (p. 286)

$$d = 2\,(2.5\cdot 10^5)\sqrt{\frac{80\,\varepsilon_0}{\mu_0}} \approx 1.2\cdot 10^4\ \text{m}.$$

(b) good conductor $\sigma \gg \varepsilon\omega$

$$\Rightarrow \varkappa \cong \omega\sqrt{\frac{\varepsilon\mu}{2}}\left(\frac{\sigma}{\varepsilon\omega}\right)^{1/2} = \sqrt{\frac{\omega\mu\sigma}{2}}$$

$$d = \sqrt{\frac{2}{\omega\mu\sigma}} \approx 1.3\cdot 10^{-8}\ \text{m} \quad \text{for } \sigma = 10^7\,\frac{1}{\Omega\cdot\text{m}}$$

$\omega \approx 10^{15}\ s^{-1}$

$\mu = \mu_0 = 4\pi\cdot 10^{-7}$

(c) Real part of $\tilde{k}$ is $k = \omega\sqrt{\frac{\varepsilon\mu}{2}}\left(\sqrt{1+\left(\frac{\sigma}{\varepsilon\omega}\right)^2}+1\right)^{1/2} \approx$

$$\approx \sqrt{\frac{\omega\mu\sigma}{2}} \quad \text{for } \sigma \gg \varepsilon\omega$$

$\Rightarrow$ eq. 9.134: $\phi = \tan^{-1}\left(\frac{\varkappa}{k}\right) = \tan^{-1}(1) = \frac{\pi}{4}$.

$$\frac{B_0}{E_0} = \left[\varepsilon\mu\sqrt{1+\left(\frac{\sigma}{\varepsilon\omega}\right)^2}\right]^{1/2} \approx \left(\varepsilon\mu\,\frac{\sigma}{\varepsilon\omega}\right)^{1/2} = \sqrt{\frac{\sigma\mu}{\omega}} \approx 10^{-7}$$

while in vacuum $\frac{B_0}{E_0} = \frac{1}{c} \approx 3\cdot 10^{-9}\ s/m$

$\Rightarrow$ mag. field is relatively stronger in metals.